

Towards Guided-Breath as an Intervention Within Computer-Assisted Learning

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ABSTRACT

Mindfulness has been actively investigated in HCI. While its positive effects on our cognitive functions are reported, carrying out mindfulness activities, in general, requires self-discipline to get its reported benefits. Our study aims to integrate a visual guide of breathing into computer-assisted learning to help the learner achieve better performance. We present within-subject research with 10 participants who learned sets of Croatian vocabulary with a multiple-choice interface showing either for 10 seconds: (a) guided-breath or (b) break instruction after every question. Our hypothesis was that (a) brings a better learner's score than (b). However, statistical analysis of learners' quiz scores did not support our hypothesis, which brought us two lessons for future improvements of our work-in-progress research. Such intervention (i) could lead to distraction rather than concentration without careful quantity and quality, and (ii) might need personalization depending on individual mindfulness levels.

CCS CONCEPTS

• Human-centered computing → Interaction design.

KEYWORDS

Mindfulness; Attention Regulation; Guided-Breath

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1 INTRODUCTION

As the increasing number of papers suggest, mindfulness has been actively investigated in HCI [6]. Some researchers work on assisting our deep breathing [3] [4] via haptic/audio/visual interface as mindfulness intervention. Off-the-shelf wearable devices also have such guided-breath^{1,2}. Such intervention through technology is helpful since it requires no instructors or self-discipline to mediate. While such work often focuses on stress-reduction [3] or assisting meditation itself [4], its usefulness within computer-assisted learning has room for exploration. The idea behind this is that mindfulness is also said to have benefits regarding academic performance [2]. In our study, guided-breath did not improve learners' performances. However, our attempts provide lessons regarding intervention design with this paper.

2 EXPERIMENTAL SETUP

Quiz Interface Design and Its Scores

We developed our software where guided-breath (GB) is embedded for our study. It displays one English word and four choices of Croatian words with only one correct option. It also has a skip button to prevent learners from selecting answers randomly when they give up. Our software picks up three distractors from the distractors pool, choosing similar ones to the correct choice using the edit distance, which requires learners to remember vocabulary clearly enough to get a question right. Our interface requires learners to try the problem and review its answers afterward.

We prepared two types of Croatian word groups in advance: A (with GB) and B (without GB). Participants learned by heart two word groups alternately to minimize the order effect, leading 6 sessions for each person. The very first word group was also randomly chosen for each person. One word group — consisting of 35 words — challenges learners with all of words in a session. The number of questions they get right during one session provides a quiz score for the session. During sessions of a specific word group, the order of quiz are randomized to expose each vocabulary evenly.

Intervention by Guided-Breath

Our 10 second GB animation is shown in Figure 1. There is a black balloon with three instructions: *inhale*, *hold*, and *exhale* in Japanese.

¹<https://support.apple.com/guide/watch/practice-mindfulness-apd371dfe3d7/>

²<https://www.mi.com/global/mi-smart-band-5/>

Table 1: Average scores over Trial-2 and Trial-3, and p-values obtained by Wilcoxon Signed-Rank test.

Trial	Average Score (A)	Average Score (B)	P-value
2	19.4	23.3	0.019
3	24.9	26.8	0.15

Following the guidance, users are supposed to breathe for 10 seconds. When (i) *inhale* is shown, the black balloon starts expanding. After that, the balloon gradually stops expanding while displaying (ii) *hold* until it gets the maximum size in the interface. Finally, the balloon starts shrinking with (iii) *exhale* until it disappears.

**Figure 1: Our guided-breath with three instructions.**

Participants & Procedure

We recruited 10 Japanese university students (4 female). Those who finished the study were compensated with gift cards of JPY 2,000. In Japan, the Croatian language is not generally spoken; We assumed participants had almost no prior knowledge of Croatian words, which was also validated by our post-experiment questionnaire.

We invited participants to a quiet room and told them to follow the instructions displayed on the screens of 23 inch desktop computers during sessions. After the briefing, we let them start using our software. An experiment for one participant is expected to be within 90 minutes. After the experiment, we asked them to answer our questionnaires. They include Five Facet Mindfulness Questionnaire (FFMQ) [1], which conceptualizes mindfulness into 5 facets, and provides scores for each facet. We used Japanese version of FFMQ [5]. Although its details cannot be described in this paper, it was used to see individual differences among participants from the mindfulness point of view.

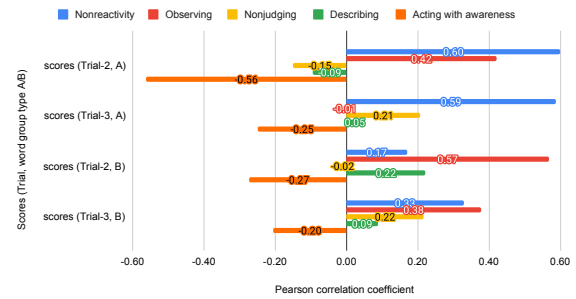
3 RESULTS & CONCLUSION

Average quiz scores of the second and the third trials are shown in Table 1. We ignored the scores of the first trials as they were only warm-ups; Since our software allows users to review answers only after they have chosen options, the quiz scores start reflecting participants' memory abilities from the second trials. The average values in Table 1 suggest that scores of group B (*without* GB) outweigh those of group A. To analyze the counter-intuitive result, we carried out Wilcoxon Signed-Rank Test (significance levels $\alpha = 0.05$) to see the performance difference between the two types of interventions. Concretely, we compared (i) the scores of group A and group B in the second trials and (ii) the scores of the two groups in the third trials. Our alternative hypothesis was that the GB led to worse results than break. P-values obtained by comparisons of group A and B are shown in Table 1. The p-value of (i) suggests that GB led to worse performance in our conditions with statistical significance, while that of (ii) cannot reject our null hypothesis.

Correlation coefficients of FFMQ facets and quiz scores over different trials are shown in Figure 2. Word group A's scores and Nonreactivity scores are positively correlated. Considering that word group A involved with GB and group A's quiz scores were

worse, GB intervention might have been distracting. The positive correlation suggests that those who had higher quiz scores in such distracting situations might have better skills of Nonreactivity. In other words, those with lower Nonreactivity might have been sensitive to GB intervention during the learning session, leading to lower quiz scores. Also, group B's scores correlate with the Observing score. Finally, only the Acting with awareness score has a negative correlation coefficient. The negative correlation coefficient is especially salient in scores (Trial-2, A).

In conclusion, our study suggests that people with different levels of mindfulness should be treated differently with care when we use GB within computer-assisted learning. Otherwise it could bring distraction rather than concentration. As future work, using today's sensing technology to choose appropriate timing/amount/frequency of GB is promising. For example, glass devices tracking our concentration levels using sensors [7] might realize *on-demand* GB intervention.

**Figure 2: Correlation between test scores and FFMQ scores.**

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REFERENCES

- [1] R. Baer, G. Smith, J. Hopkins, J. Krietemeyer, and L. Toney. 2006. Using Self-Report Assessment Methods to Explore Facets of Mindfulness. *Assessment* 13, 1 (2006), 27–45. <https://doi.org/10.1177/1073191105283504>
- [2] M. Mrazek, M. Franklin, D. Phillips, B. Baird, and J. Schooler. 2013. Mindfulness Training Improves Working Memory Capacity and GRE Performance While Reducing Mind Wandering. *Psychological Science* 24, 5 (2013), 776–781. <https://doi.org/10.1177/0956797612459659>
- [3] P. Paredes, Y. Zhou, N. Hamdan, S. Balters, E. Murnane, W. Ju, and J. Landay. 2018. Just Breathe: In-Car Interventions for Guided Slow Breathing. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 2, 1, Article 28 (mar 2018), 23 pages. <https://doi.org/10.1145/3191760>
- [4] A. Shamekhi and T. Bickmore. 2018. Breathe Deep: A Breath-Sensitive Interactive Meditation Coach. In *Proceedings of the 12th EAI International Conference on Pervasive Computing Technologies for Healthcare* (New York, NY, USA) (*PervasiveHealth '18*). Association for Computing Machinery, New York, NY, USA, 108–117. <https://doi.org/10.1145/3240925.3240940>
- [5] Y. Sugiura, A. Sato, Y. Ito, and H. Murakami. 2012. Development and Validation of the Japanese Version of the Five Facet Mindfulness Questionnaire. *Mindfulness* 12 (06 2012), 85–94. <https://doi.org/10.1007/s12671-011-0082-1>
- [6] N. Terzimehić, R. Häußelschmid, H. Hussmann, and m.c. schraefel. 2019. A Review and Analysis of Mindfulness Research in HCI: Framing Current Lines of Research and Future Opportunities. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (*CHI '19*). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3290605.3300687>
- [7] Y. Uema and K. Inoue. 2017. JINS MEME Algorithm for Estimation and Tracking of Concentration of Users. In *Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers*. Association for Computing Machinery, 297–300. <https://doi.org/10.1145/3123024.3123189>